USN | 4 | C | B | | | C | S | | | | Course Code: **18CS34**

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING, CANARA ENGINEERING COLLEGE

Programme: **B.E. (CS&E)** Semester: **III** **CIE: TEST - I** Month: **September 2019**

Course Title: **COMPUTER ORGANISATION**

Duration: **1 HOUR 30 MINUTE** Max. Marks: **50**

Note: *Answer any **ONE** full question from each part*

PART - A

1. a	Explain the working of different registers of a processor with a neat diagram.	**9**
b	With diagram explain single bus structure. Also mention at least two advantages and disadvantages of single bus structure.	**6**
c	Define clock cycle and clock rate. Explain basic performance equation.	**5**
d	How performance of a computer can be measured using SPEC rating? Explain.	**5**
e	What are Little Endian and Big Endian assignment? Explain with example and relevant diagrams.	**6**
f	Explain byte addressability and world aligned memory with diagram.	**4**

OR

2. a	Explain all four condition code flags.	**4**
b	Explain following addressing modes with suitable examples. i. Immediate ii. Register iii. Direct iv. Indirect v. Index	**10**
c	Explain how stack works with diagram. Write simple assembly routines for implementing operation of safe push and safe pop operations with full and empty errors respectively.	**9**
d	Explain all rotate instructions with diagram and example.	**12**

PART - B

3. a	With diagram explain parts of hardware required to connect an I/O device to the bus.	**7**
b	What is the use of DATAIN, DATAOUT, STATUS and CONTROL registers in I/O interface of keyboard and display? Also write an assembly language routine that reads one line from keyboard stores it in memory buffer and echoes it back to display.	**8**

OR

4. a	What are interrupts, Interrupt Service Routines and interrupt latency?	**3**
b	List the sequence of events involved in handling an interrupt request from a single device.	**6**
c	Explain daisy chain and priority groups with diagram and at least one advantage.	**6**

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